

CHAPTER 10

How Quantities Combine: Understanding Data

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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🤔 Think and Reflect · 10.1.1 Average of Averages

? **Badminton academy: 8 seniors (average 165.5 cm) and 3 juniors (average 149.33 cm). Method 1: (165.5 + 149.33)/2 = 157.415. Method 2: sum of all 11 heights ÷ 11 = 161.09. Whose calculation gives the correct average height of the whole group?**

✔ ANSWER

Average of the whole group = (sum of all 11 heights) ÷ 11.

So **Method 2 is correct: 161.09 cm**.

1 **Why did Method 1 not work?**

✔ ANSWER

Method 1 gives the seniors' average and the juniors' average equal importance.

But there are 8 seniors and only 3 juniors, so the seniors' average should count more.

Method 1 ignores the different group sizes.

? **Shreyas found (165.5 × 8 + 149.33 × 3)/(8 + 3) = 1772/11 = 161.09. Do you understand why this also works? Why does 165.5 × 8 give the sum of the heights of all the seniors?**

✔ ANSWER

Average = sum ÷ number, so sum = average × number.

Sum of seniors' heights = 165.5 × 8 = 1324; sum of juniors' heights = 149.33 × 3 = 448.

Adding gives the sum of all 11 heights (1772); dividing by 11 gives **161.09 cm**, the same as Method 2.

? **Jaspreet cycled 5 days each week for 3 weeks (weekly averages 12.8, 15.8, 18 minutes). Why does Method 1 (average of the 3 weekly averages) give the correct answer here? When does it work and when does it not?**

✔ ANSWER

Each week has the same number of days (5), so each weekly average comes from the same number of values.

Then (5a + 5b + 5c)/(5 + 5 + 5) = (a + b + c)/3.

Averaging the averages works only when all the collections have the same size. If the sizes differ, we must use the weighted mean.

❓ In-text question · 10.1.2 Mixtures

? **500 mL lemonade with 10% jaggery is mixed with 200 mL lemonade with 20% jaggery. Can you estimate what part of the mixture is jaggery? Is it 15%, or more or less? Why?**

✔ ANSWER

It is **less than 15%**, closer to 10%, because more lemonade comes from the 10% bowl.

Check: (500 × 0.1 + 200 × 0.2)/(500 + 200) = 90/700 ≈ **0.13 = 13%**.

📝 Exercise Set 10.1

1 **The average score of students on a test in Section A is 72 and that of Section B is 76. What is the combined average if Section A has 30 students and Section B has 25 students?**

✔ ANSWER

Sum of scores in A = 72 × 30 = 2160

Sum of scores in B = 76 × 25 = 1900

Combined average = (2160 + 1900)/(30 + 25)

= 4060/55

= **73.82 (approx.)**

2 **A farmer mixes three equal quantities of fertilisers. The first contains 1/10 nitrogen, the second 9/50 nitrogen and the third 3/60 nitrogen. What is the fraction of nitrogen in the mixture?**

✔ ANSWER

Equal quantities, so the simple average of the fractions works.

1/10 = 30/300, 9/50 = 54/300, 3/60 = 15/300

Sum = 99/300

Fraction in mixture = (99/300) ÷ 3 = 99/900

= **11/100** (that is, 11%)

3 **(Śrīdharācārya, Pāṭīgaṇita) A goldsmith melts together 9 units of gold at 12 varṇa, 5 units at 10 varṇa and 17 units at 11 varṇa. Find the purity in varṇa of the combined gold.**

✔ ANSWER

Purity = (9 × 12 + 5 × 10 + 17 × 11)/(9 + 5 + 17)

= (108 + 50 + 187)/31

= 345/31

= **11 4/31 varṇa ≈ 11.13 varṇa**

4 **The average rainfall per day in May, June and July at a place is 3.5 mm, 10 mm and 8.7 mm respectively. Write an expression that gives their combined average.**

✔ ANSWER

May has 31 days, June 30 days, July 31 days. So the weights are 31, 30, 31.

Combined average = (3.5 × 31 + 10 × 30 + 8.7 × 31)/(31 + 30 + 31)

= (108.5 + 300 + 269.7)/92 = 678.2/92 ≈ 7.37 mm per day

5 (i) **A 100 mL kashayam/kadha with 5% spice mix, a 200 mL one with 10% and a 300 mL one with 15% are combined. Find the concentration of spice mix.**

✔ ANSWER

Spice = 100 × 0.05 + 200 × 0.10 + 300 × 0.15 = 5 + 20 + 45 = 70 mL

Total = 100 + 200 + 300 = 600 mL

Concentration = 70/600 = **11.67% (approx.)**

5 (ii) **A 300 mL kadha with 5% spice mix, a 200 mL one with 10% and a 100 mL one with 15% are mixed. Find the concentration.**

✔ ANSWER

Spice = 300 × 0.05 + 200 × 0.10 + 100 × 0.15 = 15 + 20 + 15 = 50 mL

Total = 600 mL

Concentration = 50/600 = **8.33% (approx.)**

(More of the weak kadha here, so the mixture is weaker than in (i).)

📝 Exercise Set 10.2

1 **Savitri's marks in Kashmiri: 35 out of 50 (internal tests), 44 out of 60 (project), 80 out of 100 (final exam). They are combined in the ratio 3 : 4 : 5. Which expression(s) give her annual score as a percentage?**

(i) (35 × 3 + 44 × 4 + 80 × 5)/(3 + 4 + 5)

(ii) ((35/100) × 3 + (44/100) × 4 + (80/100) × 5)/(3 + 4 + 5)

(iii) [((35/50) × 3 + (44/60) × 4 + (80/100) × 5)/(3 + 4 + 5)] × 100

(iv) ((35/50) × 100 × 3 + (44/60) × 100 × 4 + (80/100) × 100 × 5)/(3 + 4 + 5)

✔ ANSWER

Each mark must first be changed to a fraction of its own maximum (35/50, 44/60, 80/100).

(i) uses raw marks with different maximums – wrong.

(ii) divides 35 and 44 by 100 instead of 50 and 60 – wrong.

(iii) and (iv) are the same expression (×100 taken outside or inside) – correct.

Answer: (iii) and (iv).

Her score = (0.7 × 3 + 0.733 × 4 + 0.8 × 5)/12 × 100 = 9.033/12 × 100 ≈ **75.28%**

📝 Exercise Set 10.3

1 **A stationery shop owner made ₹8000 selling books, of which 30% is profit, and ₹1000 selling book covers, of which 50% is profit. What is the percentage of profit on the total sales?**

✔ ANSWER

Profit on books = 30% of 8000 = ₹2400

Profit on covers = 50% of 1000 = ₹500

Total profit = ₹2900 on total sales ₹9000

Profit % = 2900/9000 × 100 = **32.22% (approx.)**

2 **A white stork's average daily distance over 20 days is 44.5 km. On the 21st day it flew 55 km. What is the average daily distance over these 21 days? Make a guess before you calculate.**

✔ ANSWER

Guess: a little more than 44.5 km, since only one day is added.

Distance in 20 days = 44.5 × 20 = 890 km

Total in 21 days = 890 + 55 = 945 km

Average = 945/21 = **45 km**

3 **A 600 mL solution with 5% salt is mixed with a 300 mL solution with 8% sugar. What are the concentrations of salt and sugar in the mixture?**

(i) Salt 5%, Sugar 8% (ii) Salt 13%, Sugar 3% (iii) Salt 6%, Sugar 6% (iv) Salt 5.55%, Sugar 8.88% (v) Salt 3.33%, Sugar 2.67% (vi) Salt 4.1%, Sugar 7.08%

✔ ANSWER

Total mixture = 600 + 300 = 900 mL

Salt = 5% of 600 = 30 → 30/900 = 3.33%

Sugar = 8% of 300 = 24 → 24/900 = 2.67%

Option (v): Salt 3.33%, Sugar 2.67%

4 **The concentration of spice in 10 litres of pani is 8%. What quantity of regular water should be mixed so that the spice level is reduced to 3/4 of the original concentration?**

✔ ANSWER

New concentration = 3/4 × 8% = 6%

Spice in pani = 8% of 10 L = 0.8 L (does not change)

Let w litres of water be added: 0.8/(10 + w) = 0.06

10 + w = 0.8/0.06 = 13.33

w = **3 1/3 litres (about 3.33 L)**

5 (i) **Final marks combine strength, flexibility and agility in the ratio 4 : 5 : 6. Rashi scored 60, 65, 70; Keerthi scored 55, 65, 75. Find out whose total is more without calculating.**

✔ ANSWER

Flexibility is the same (65).

Keerthi has 5 less in strength (weight 4) but 5 more in agility (weight 6, a bigger weight).

So **Keerthi's final marks are more**.

5 (ii) **What are their final marks?**

✔ ANSWER

Rashi = (60 × 4 + 65 × 5 + 70 × 6)/15 = (240 + 325 + 420)/15 = 985/15 ≈ **65.67**

Keerthi = (55 × 4 + 65 × 5 + 75 × 6)/15 = (220 + 325 + 450)/15 = 995/15 ≈ **66.33**

6 **10 customers rated a restaurant from 1 to 5 (number of customers giving each rating):**

Rating	5	4	3	2	1
Food	5	3	2	0	0
Ambience	0	4	5	1	0
Service	1	2	2	4	1

What is the average rating if food : ambience : service = 6 : 5 : 4?

✔ ANSWER

Food = (5×5 + 4×3 + 3×2)/10 = 43/10 = 4.3

Ambience = (4×4 + 3×5 + 2×1)/10 = 33/10 = 3.3

Service = (5×1 + 4×2 + 3×2 + 2×4 + 1×1)/10 = 28/10 = 2.8

Combined = (6 × 4.3 + 5 × 3.3 + 4 × 2.8)/(6 + 5 + 4)

= (25.8 + 16.5 + 11.2)/15 = 53.5/15 ≈ **3.57**

7 **Langurs: average weight of males 16.5 kg, females 13.8 kg, all 60 langurs 14.925 kg. Without computing, are there more males or more females, or equal?**

✔ ANSWER

If numbers were equal, the overall average would be midway: (16.5 + 13.8)/2 = 15.15 kg.

14.925 is less than 15.15, i.e. closer to the female average.

There are more female langurs.

7 (i) **Which expression(s) describe the scenario (x males, y females)?**

(a) (16.5x + 13.8y)/(x + y) = 14.925

(b) (16.5x + 13.8y)/60 = 14.925

(c) (16.5x + 13.8y)/2 = 14.925

(d) (16.5x + 13.8y)/(16.5 + 13.8) = 14.925

✔ ANSWER

Total weight ÷ total number = overall average, and x + y = 60.

(a) and (b) are correct. (c) and (d) divide by the wrong number.

7 (ii) **Find out how many male langurs are present.**

✔ ANSWER

Let x = males, so females = 60 − x.

16.5x + 13.8(60 − x) = 14.925 × 60

16.5x + 828 − 13.8x = 895.5

2.7x = 67.5

x = **25 male langurs** (and 35 females)

7 (iii) **A female langur weighing 15.2 kg is admitted. What is the average weight of the female langurs after this?**

✔ ANSWER

Total weight of 35 females = 35 × 13.8 = 483 kg

New total = 483 + 15.2 = 498.2 kg for 36 females

Average = 498.2/36 ≈ **13.84 kg**

7 (iv) **Two male langurs weighing 16.9 kg and 16.1 kg are released. What is the average weight of the male langurs after this?**

✔ ANSWER

Total weight of 25 males = 25 × 16.5 = 412.5 kg

Remove 16.9 + 16.1 = 33 kg → 379.5 kg for 23 males

Average = 379.5/23 = **16.5 kg** (unchanged, since the two removed langurs average 16.5 kg)

7 (v) **Now one of the male langurs loses 1 kg. What is the average weight of all the male langurs after this?**

✔ ANSWER

Total = 379.5 − 1 = 378.5 kg for 23 males

Average = 378.5/23 ≈ **16.46 kg**

(The average falls by 1/23 kg.)

8 (i) **Salinity: Dead Sea ≈ 34%, other seas ≈ 3.5%, ground water ≈ 0.01%, purified drinking water ≈ 0.001%.**

What is the salinity if 1 litre of Dead Sea water is mixed with 2 litres of purified drinking water?

✔ ANSWER

Salinity = (1 × 34 + 2 × 0.001)/(1 + 2)

= 34.002/3

≈ **11.33%**

8 (ii) **Is it possible to mix Dead Sea water and purified drinking water to get the salinity of groundwater? Why/why not? How much purified water should be mixed with 1 litre of Dead Sea water?**

✔ ANSWER

Yes. A mixture's salinity always lies between the two salinities, and 0.01% lies between 0.001% and 34%.

Let w litres of purified water be added:

(34 + 0.001w)/(1 + w) = 0.01

34 + 0.001w = 0.01 + 0.01w

33.99 = 0.009w

w = 33.99/0.009 ≈ **3777 litres** (about 3,777 L)

8 (iii) **Is it possible to mix Dead Sea water and groundwater to get the salinity of purified drinking water? Why/why not? How much groundwater is needed?**

✔ ANSWER

No. The mixture's salinity must lie between 0.01% and 34%.

Purified water (0.001%) is less salty than both, so it can never be reached, however much groundwater is added. No quantity works.

❓ Questions · 10.2.1 Stacking Columns (Table 10.2)

Data Average weekly expenditure (₹) of three families:

Family	Housing	Food	Education	Transportation	Healthcare	Recreation
A	1940	1700	1280	1200	1770	535
B	1750	1546	1500	1280	1210	0
C	950	1700	1540	1400	1300	150

✔ ANSWER

Answers to Questions 1–8 below use this table (Fig. 10.1 and 10.2 show the same data).

1 Which family spent the least on housing?

✔ ANSWER

Family C (₹950).

2 Which family spent the most on healthcare?

✔ ANSWER

Family A (₹1770).

3 Approximately how much did Family A spend on education?

✔ ANSWER

About ₹1300 (exactly ₹1280).

4 Which category did Family B spend most on?

✔ ANSWER

Housing (₹1750).

5 Did Family A spend more on food or on healthcare?

✔ ANSWER

Healthcare (₹1770), slightly more than food (₹1700).

6 Which family spent the least amount overall?

✔ ANSWER

A: 1940 + 1700 + 1280 + 1200 + 1770 + 535 = ₹8425

B: 1750 + 1546 + 1500 + 1280 + 1210 + 0 = ₹7286

C: 950 + 1700 + 1540 + 1400 + 1300 + 150 = ₹7040

Family C spent the least.

7 **Approximately how much did Family C spend overall?**

✔ ANSWER

About ₹7000 (exactly ₹7040).

8 **Approximately what percentage of Family C's expenses were on housing?**

✔ ANSWER

950/7040 × 100 ≈ **13.5%**

? **Use the stacked bar chart (Fig. 10.2) to answer Questions 1–5. Is it possible? Is it easy? If not, what made it harder?**

✔ ANSWER

It is possible, but harder.

Except the first part, the small bars do not start from 0, so their sizes are hard to read.

Bars of the same category in different families do not start at the same point, so comparing them is difficult. Totals (Q 6–8) are easier to see.

❓ In-text questions · 10.2.2 An Alternative to Pie Chart

? **Which family has the smallest share of their total expenditure towards healthcare? Was it easy from the stacked bar chart?**

✔ ANSWER

A: 1770/8425 ≈ 21.0%; B: 1210/7286 ≈ 16.6%; C: 1300/7040 ≈ 18.5%

Family B has the smallest share.

It is not easy from the stacked bar chart; pie charts or a 100% stacked bar chart show shares more clearly.

Ex 7 **Daily electricity use of a house (kWh):**

Year	Lighting	Cooling	Kitchen appliances	Other appliances
2005	360	300	120	220
2025	220	800	460	520

Make the 100% stacked bar chart. What do you observe? What do you find interesting?

💡 SAMPLE ANSWER

2005: total 1000 → Lighting 36%, Cooling 30%, Kitchen 12%, Other 22%.

2025: total 2000 → Lighting 11%, Cooling 40%, Kitchen 23%, Other 26%.

Draw each year as a bar of the same length divided in these percentages.

Observations: total use doubled; lighting use fell (maybe LED bulbs) while cooling became the biggest share; kitchen and other appliances grew a lot.

Ex 8 **100% stacked bars: Fatima – Summer 50%, Monsoon 30%, Winter 20%; Naveen – Summer 40%, Monsoon 25%, Winter 35%. Which can be inferred?**

(i) In Fatima's garden, more blooms in summer than in monsoon.

(ii) In summer, Fatima's garden had more blooms than Naveen's.

(iii) In winter, Fatima's garden had fewer blooms than Naveen's.

(iv) The total number of flowers is the same in both gardens.

✔ ANSWER

Only (i) can be inferred: it compares parts of the same garden's total.

(ii), (iii), (iv) cannot: the chart does not show how many flowers each garden had in total, so actual numbers cannot be compared.

🤔 Think and Reflect · What Can We Infer?

1 **What does this say about the scope of stacked bars and 100% stacked bars?**

✔ ANSWER

A stacked bar shows actual amounts, so we can compare totals and parts across bars.

A 100% stacked bar shows only shares (percentages); it cannot tell us actual numbers or compare amounts across bars.

2 **Given a stacked bar chart, can we make a corresponding 100% stacked bar chart?**

✔ ANSWER

Yes. Divide each part by its bar's total and multiply by 100.

3 **Given a 100% stacked bar chart, can we make a corresponding stacked bar chart?**

✔ ANSWER

No (not uniquely). The totals are not known; many different totals give the same percentages (e.g. Case 1 and Case 2).

4 **What kind of inferences or comparisons can be made from a stacked bar chart and a 100% stacked bar chart?**

✔ ANSWER

Stacked bar: compare totals, and estimate actual amounts of parts.

100% stacked bar: compare the shares (fractions) of parts within a bar, and compare shares across bars – but not actual amounts.

⭐ A Typical Day (Example 9)

Ex 9 **The chart shows the time spent on activities in a day by the average Indian: Sleep 34%, Personal care 13%, Paid work 10%, Unpaid work & care 11%, Learning 5%, Leisure, social & travel 27%. What do you notice? What do you wonder about?**

💡 SAMPLE ANSWER

Notice: about one-third of the day (about 8 h) goes in sleep; leisure, social and travel take more than a quarter (about 6.5 h); learning gets only about 5% (about 1.2 h).

Wonder: how does this differ for children, adults and the elderly, for men and women, or for villages and cities?

🤔 Think and Reflect · Time use by age group (Fig. 10.6)

1 **What do you find interesting in the chart? What can you infer? Discuss.**

💡 SAMPLE ANSWER

Children (6–14) spend about 22% of the day learning; adults (25–59) almost none.

Adults spend the most on paid work (14%) and unpaid work & care (15%).

The elderly have the most leisure (33%) and sleep more than adults.

Personal care is about the same (13–14%) for all ages.

2 **Do you remember the sleep time over age trend you studied last year? Does it align with this chart?**

💡 SAMPLE ANSWER

Children sleep the most (38%, about 9 h); sleep falls for youth (34%) and adults (32%, about 7.7 h), and rises again for the elderly (37%).

This matches the usual trend that sleep need decreases from childhood to adulthood.

3 **Why has the learning time of the 15–24 age group reduced significantly compared to that of 6–14?**

💡 SAMPLE ANSWER

Many people of 15–24 have finished school or college and started paid work (7%) or household work and care (9%).

Some leave studies early to earn for the family.

4 **Do adults spend about an equal amount of time in paid and unpaid work?**

💡 SAMPLE ANSWER

Yes, roughly: paid work about 14% (≈ 3.4 h) and unpaid work & care about 15% (≈ 3.6 h).

But this is an average: some adults (often men) do mostly paid work and others (often women) do mostly unpaid work.

🤔 Think and Reflect · 'What Can A Strip Say?'

? **In each strip, if we club the tiny strips belonging to each activity together, will we get a 100% stacked bar chart like the one in Fig. 10.6?**

✔ ANSWER

Yes. Each strip covers the whole day (24 h). Joining the pieces of each activity gives one part per activity, and the strip becomes a 100% stacked bar (and also a stacked bar, since every total is 24 h).

Exercise Set 10.4

1 (i) **Stacked column chart: number of animal species on the IUCN Red List by class (2007: 7,851; 2010: 9,618; 2013: 11,212; 2016: 12,630; 2019: 14,234). What does the number 14,234 on top of the 2019 column represent?**

✔ ANSWER

It is the **total number of animal species (all classes together) on the IUCN Red List in 2019**.

1 (ii) **Approximately how many reptile species were on the list in 2016?**

💡 SAMPLE ANSWER

The reptile (dark red) part of the 2016 column goes from about 1,000 to about 2,100.

So about **1,100 reptile species** (check the picture in your book).

1 (iii) **Which class(es) of species have seen a relatively small increase in count between 2007 and 2019?**

💡 SAMPLE ANSWER

The parts that hardly changed in height: **Mammals** (about 1,000 → about 1,200), and **Birds** grew only a little.

Insects, fish, molluscs and amphibians grew much more. (check the picture in your book)

2 (i) **Wickets taken by a bowler:**

Place	Test	ODI	T20
Home	62	100	32
Overseas	172	49	71

Complete the stacked bar chart comparing total wickets at home vs. overseas.

✔ ANSWER

Home bar: Test 0 to 62, ODI 62 to 162, T20 162 to **194** (total).

Overseas bar: Test 0 to 172, ODI 172 to 221, T20 221 to **292** (total).

The overseas bar is clearly longer.

2 (ii) **Complete the stacked bar chart showing the total wickets taken in each format (parts: Home, Overseas).**

✔ ANSWER

Test bar: Home 0 to 62, Overseas 62 to **234**.

ODI bar: Home 0 to 100, Overseas 100 to **149**.
T20 bar: Home 0 to 32, Overseas 32 to **103**.

📝 Exercise Set 10.5

1 (i) **Average hours per day (Mon–Fri):**

Person	Lying down	Sitting	Standing/moving
Pavani (Patient)	20	3	1
Raghu (Nurse)	7	4	13
Zakir (Teacher)	8	6	10
Sahana (Student)	?	?	?
Julie (____)	8	10	6

How much time does Sahana spend in each body-state? Make a reasonable guess.

💡 SAMPLE ANSWER

A student sleeps about 8–9 h and sits in class and while studying.
Sample: **Lying down 9 h, Sitting 9 h, Standing/moving 6 h** (total 24 h).

1 (ii) **Guess what activity or work Julie could be engaged in, based on the data (8 h lying, 10 h sitting, 6 h standing/moving).**

💡 SAMPLE ANSWER

She sits for most of the waking day, so she probably has a desk job, e.g. **office worker, computer operator, bank clerk or driver**.

1 (iii) **Complete the 100% stacked bar chart based on the table.**

💡 SAMPLE ANSWER

Each total is 24 h, so % = hours/24 × 100:

Person	Lying down	Sitting	Standing/moving
Pavani	83.3%	12.5%	4.2%
Raghu	29.2%	16.7%	54.2%
Zakir	33.3%	25%	41.7%
Sahana (sample 9, 9, 6)	37.5%	37.5%	25%
Julie	33.3%	41.7%	25%

Draw each person's bar of the same length divided in these percentages.

🤔 Think and Reflect · Exercise Set 10.5

1 **Will the bars look similar if a different teacher's data is considered instead of Zakir's?**

💡 SAMPLE ANSWER

Roughly similar, since teachers stand and move a lot in class, but not the same: it depends on the subject, travel, and how much time is spent sitting for checking copies or planning.

2 **Will the bars look different if the data for all 7 days of the week is considered?**

💡 SAMPLE ANSWER

Yes, probably. On Saturday and Sunday many people rest more and work less, so lying down and sitting may increase for the teacher, nurse and student. The patient's bar may stay almost the same.

3 **Some bars may also match people doing other kinds of work/activities. Can you think of any?**

💡 SAMPLE ANSWER

Pavani's bar: any person on bed rest or a small baby.

Raghu's bar: a shopkeeper, waiter, traffic police or farmer.

Julie's bar: an office worker, bus driver or tailor.

📝 End-of-Chapter Exercises

1 (i) **Run rate = average runs per over. A team scored 6 runs in the first over (run rate 6). In the second over they scored 12 runs. What is the run rate now?**

✔ ANSWER

Runs = 6 + 12 = 18 in 2 overs

Run rate = 18/2 = **9**

1 (ii) **After Over 19 their run rate was 6. What is the run rate after 20 overs if they made 12 runs in the last over?**

✔ ANSWER

Runs in 19 overs = 6 × 19 = 114

Runs in 20 overs = 114 + 12 = 126

Run rate = 126/20 = **6.3**

2 **Shuttlecock prices (₹) collected by five friends (N = nylon, F = feather):**

Yusuf: 40(N), 105(N), 383(F), 108(N), 165(F), 116(F)

Srikanth: 194(N), 85(N), 93(N), 121(N)

Kashvi: 49(N), 297(F), 105(N), 275(F), 40(N)

Prasanna: 124(N), 333(F), 182(N), 258(F)

Gracy: 220(F), 458(F), 129(F), 183(N)

Can you describe a way they can work together to find the average price of a shuttlecock across types?

✔ ANSWER

Each friend finds the sum (or average) of his/her own prices and counts them. Add all the sums and divide by the total count (6 + 4 + 5 + 4 + 4 = 23).

Total = 917 + 493 + 766 + 897 + 990 = 4063

Average = 4063/23 ≈ **₹176.65**

2 (i) **Each one's average is ay, as, ak, ap, ag. Write an expression for the combined average.**

✔ ANSWER

Weights = number of prices each one collected (6, 4, 5, 4, 4).

Combined average = (6ay + 4as + 5ak + 4ap + 4ag)/23

2 (ii) **If an, af are the average prices of nylon and feather shuttlecocks, write an expression for the average price of a shuttlecock. Will it equal the answer of part (i)?**

✔ ANSWER

Nylon: 3 + 4 + 3 + 2 + 1 = 13 prices; Feather: 3 + 0 + 2 + 2 + 3 = 10 prices.

Average = (13an + 10af)/23

Yes, it is equal: both give (sum of all 23 prices)/23.

Check: an = 1429/13 ≈ 109.92, af = 2634/10 = 263.4; (1429 + 2634)/23 = 4063/23 ≈ ₹176.65.

3 (i) **Shreyas holds 25 shares at an average price of ₹150. He buys 10 shares at ₹30 each. What is his average price per share now?**

✔ ANSWER

Cost = 25 × 150 + 10 × 30 = 3750 + 300 = ₹4050

Shares = 35

Average = 4050/35 ≈ **₹115.71**

3 (ii) **Vaishnavi holds 5 shares at an average price of ₹150. She buys some shares at ₹30 each and her average becomes ₹70. How many shares did she buy?**

✔ ANSWER

Let n shares be bought.

$(5 \times 150 + 30n)/(5 + n) = 70$

$750 + 30n = 350 + 70n$

$400 = 40n$

$n =$ **10 shares**

4 **(Pṛthūdakasvāmī) A pool 30 hastas long has 5 sections of lengths 4, 5, 6, 7 and 8 hastas, dug to depths 9, 7, 7, 3 and 2 hastas respectively. Find the mean depth.**

✔ ANSWER

Mean depth $= (4\times9 + 5\times7 + 6\times7 + 7\times3 + 8\times2)/30$

$= (36 + 35 + 42 + 21 + 16)/30$

$= 150/30$

= **5 hastas**

5 **Suvarna has 1 g gold bought at ₹15k (point O). Find the average price after each transaction:**
(i) Buy 1 g at ₹30k (ii) Buy 2 g at ₹30k (iii) Buy 1 g at ₹10k
(iv) Buy 10 g at ₹10k (v) Buy 0.5 g at ₹30k (vi) Buy 0.5 g at ₹15k

✔ ANSWER

(i) $(15 + 30)/2 =$ **₹22.5k**

(ii) $(15 + 2 \times 30)/3 = 75/3 =$ **₹25k**

(iii) $(15 + 10)/2 =$ **₹12.5k**

(iv) $(15 + 10 \times 10)/11 = 115/11 \approx$ **₹10.45k** (very close to 10)

(v) $(15 + 0.5 \times 30)/1.5 = 30/1.5 =$ **₹20k**

(vi) **₹15k** (same price, so no change)

Mark each value on the number line; the new point always lies between 15 and the new price, closer to the one with more grams.

6 **How would the weighted average change if all the weights are doubled? Justify using algebra.**

✔ ANSWER

Example: values 10, 20 with weights 1, 3 → $(10 + 60)/4 = 17.5$; weights 2, 6 → $(20 + 120)/8 = 17.5$.

It does not change.

Proof: $(2w_1x_1 + 2w_2x_2 + \ldots + 2w_nx_n)/(2w_1 + 2w_2 + \ldots + 2w_n) = 2(w_1x_1 + \ldots + w_nx_n)/[2(w_1 + \ldots + w_n)] = (w_1x_1 + \ldots + w_nx_n)/(w_1 + \ldots + w_n)$.

Only the ratio of the weights matters.

7 (i) **Graph: cumulative number of objects orbiting Earth (payload and other objects) by year. In 2003, approximately how many total objects were there? In what year did this number double?**

💡 SAMPLE ANSWER

In 2003 there were about **10,000 objects**.

The total reached about 20,000 around **2021–2022** (check the picture in your book).

7 (ii) **Find the approximate number and share of payload objects and other objects in 2025.**

💡 SAMPLE ANSWER

Total in 2025 ≈ 33,000.

Payload objects ≈ **16,500 (about 50%)**; other objects ≈ **16,500 (about 50%)** (check the picture in your book).

7 (iii) **What can you say about the number of payload objects and other objects over time, and their shares over time?**

💡 SAMPLE ANSWER

Payload objects grew slowly for about 60 years, then shot up after about 2019 (many new satellites).

Other objects (debris, rocket stages) grew steadily, with a big jump around 2007.

Share: for most years payload objects were only about 20–25% and other objects about 75–80%. Recently the payload share rose quickly to about half, and the share of other objects fell.

8 (i) **Infographic (map): % of schools with a playground – e.g. Punjab 98%, Haryana 90%, Uttarakhand 78%, Arunachal Pradesh 68%, Maharashtra 93%, Telangana 74%. Identify the correct inference(s):**

(a) More schools have a playground in Haryana than in Uttarakhand.

(b) About 2 out of 3 schools in Arunachal Pradesh have a playground.

(c) Punjab has the highest number of schools with a playground.

(d) If Maharashtra has more schools than Telangana, then more schools have a playground in Maharashtra.

✔ ANSWER

(a) Not sure: the map gives percentages, not numbers of schools.

(b) **Correct**: 68% ≈ 2/3.

(c) Not sure: Punjab has the highest percentage, not necessarily the highest number.

(d) **Correct**: Maharashtra has more schools and a higher percentage (93% > 74%), so 93% of a bigger number is more.

Answer: (b) and (d).

8 (ii) **Using the information given, can we find the nation-wide percentage of schools with a playground? If not, what additional information is needed?**

✔ ANSWER

No. We need the **number of schools in each state/UT**.

Then the national % = weighted mean of the state percentages, with the number of schools as weights.

9 (i) **Decision Dilemma: Which play would you choose to watch based on the 100% stacked bar chart of the share of ratings (★★★★★ to ★) of Plays A, B and C?**

💡 SAMPLE ANSWER

Plays A and B have about 70% of ratings as 4 or 5 stars and few 1-star ratings.

Play C has the most 5-star ratings (about 40%) but also about one-third 1-star ratings.

So I would choose **Play A or Play B** (check the picture in your book).

9 (ii) **The corresponding stacked bar chart (number of ratings) is shown. Would you change your decision? Why/why not?**

💡 SAMPLE ANSWER

The stacked chart shows Play A has only about 200 ratings, Play C about 380, and Play B about 900.

Play B's good share comes from many more people, so it is more reliable.

So I would choose **Play B**. (A play with few ratings may look good or bad by chance.)

10 (i) **Triathlon finish times (hh:mm):**

Athlete	Swimming	Cycling	Running
1	01:08	05:00	03:15
2	01:05	05:10	03:35

3	01:22	05:55	03:50

What is a suitable representation – a stacked bar chart or a 100% stacked bar chart? Why?

✔ ANSWER

Totals: Athlete 1 = 9 h 23 min, Athlete 2 = 9 h 50 min, Athlete 3 = 11 h 07 min.

A stacked bar chart, because the total time decides the winner, and it also shows the time of each segment.

10 (ii) **If a 100% stacked bar chart is drawn, which question(s) can be answered from just that chart?**

(a) Who finished the race first?

(b) Approximately what fraction of their race time did Athlete 1 spend cycling?

(c) Which athlete took the longest for running?

✔ ANSWER

Only (b). Athlete 1 cycled 300 of 563 minutes ≈ 53%, about half.

(a) and (c) need actual times, which a 100% chart does not show.

11 **Graph: distribution of disabled persons by age group and type of disability in India (Census 2011), as 100% bars for ages 0–19, 20–39, 40–59, 60 & above. What do you notice? What do you wonder? Write your inferences.**

💡 SAMPLE ANSWER

The share of movement disability grows with age and is the largest for 60 & above.

Seeing and hearing disabilities form a big share in every age group, largest among the elderly.

Mental retardation, 'any other' and multiple disability have a bigger share among the young (0–19).

Wonder: how many people are in each group? The chart shows only shares, not numbers.

12 **Individual project: (i) record time spent lying down, sitting and standing/moving for yourself and 2 family members and draw a stacked bar chart; or (ii) show your family's monthly expenditure for at least 3 months in a 100% stacked bar chart and write observations.**

💡 SAMPLE ANSWER

(i) Sample: Me – lying 9 h, sitting 9 h, standing/moving 6 h; Mother – 7 h, 5 h, 12 h; Grandfather – 10 h, 9 h, 5 h. Draw a bar of 24 units for each person split into the three parts.

(ii) Sample categories: food, rent/housing, education, travel, health, other. For each month, % = category ÷ month total × 100; draw one 100% bar per month, then compare the shares (e.g. food has the biggest share every month).

13 **Small-group project: design a custom rating scheme with weights (cloth store, bus travel, tourist spot or clinic), collect 10 ratings, compute ratings, make a 100% stacked bar chart and write a note.**

💡 SAMPLE ANSWER

Sample (bus travel): aspects – safety, punctuality, cleanliness, comfort, fare; weights 5 : 4 : 3 : 2 : 1 (safety matters most).

One person's rating = (5×safety + 4×punctuality + 3×cleanliness + 2×comfort + 1×fare)/15.

Overall average = mean of the 10 individual ratings.

100% stacked bar: for each aspect, show the share of 5★, 4★, … ratings.

Note: weights show what matters most, but they miss things not rated (e.g. crowding).

14 **Whole class project: each student shares the average age of their family and the number of family members. Find the average age of all the families of the class.**

💡 SAMPLE ANSWER

Use the weighted mean with family size as the weight:

Average age = $(a_1n_1 + a_2n_2 + \ldots + a_kn_k)/(n_1 + n_2 + \ldots + n_k)$, where a_i = average age and n_i = members of family i.

(Simply averaging the averages is wrong unless all families have the same size.)

***15** **What happens to the weighted average if all the weights are increased by a constant value, say 1? Justify using algebra.**

✔ ANSWER

Example: values 10, 20 with weights 1, 3 → 70/4 = 17.5; weights 2, 4 → 100/6 ≈ 16.67. It changed.

New mean = $(\Sigma(w_i + 1)x_i)/(\Sigma(w_i + 1)) = (\Sigma w_ix_i + \Sigma x_i)/(\Sigma w_i + n)$.

This is in general **not equal** to $\Sigma w_ix_i/\Sigma w_i$; the weights become more equal, so the weighted mean moves **towards the simple mean** of the values.

It stays the same only in special cases, e.g. when all weights were already equal (or the weighted mean equals the simple mean).

***16** **Last year cows were 60%, sheep 25% and chickens 15% of the farm animals. The number of all three decreased. Which changes in their shares are possible?**

(i) all three % decreased

(ii) all three % increased

(iii) all three % remained the same

(iv) cows % decreased, sheep % increased, chickens % same

(v) cows % increased, sheep % increased, chickens % decreased

✔ ANSWER

Shares always add up to 100%, so they cannot all go down or all go up: (i) and (ii) are **not possible**.

(iii) Possible: e.g. 60, 25, 15 → 48, 20, 12 (all fall by 20%).

(iv) Possible: e.g. 60, 25, 15 → 50, 24, ≈13.06 (total ≈ 87.06): 57.4%, 27.6%, 15%.

(v) Possible: e.g. 60, 25, 15 → 55, 23, 5 (total 83): 66.3%, 27.7%, 6%.

Answer: (iii), (iv) and (v).

🎉 **Chapter 10 done! Revise the answers once more tomorrow.**